Architecture - Structural Correspondence with the 5-Stage Model

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Architecture

Structural correspondence survey with the 5-stage model (Ba / Wave / En / Uzu / Taba)

Survey Overview

- **Survey target**: 12 major theories in architecture
- **Research question**: Do the various theories of architecture correspond structurally to the 5-stage model?
- **Results**: Strong correspondence in 1 case, conditional correspondence in 1 case

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Overview of the 5-Stage Model

Stage	Definition
Ba (Field)	An undifferentiated state. The initial condition in which no direction or structure has yet been established
Wave	The exploratory stage in which multiple directions diverge and compete
En (Edge)	A state of tension in which opposing elements coexist without converging on either side. The place where they meet at the boundary, influence each other, and relationships emerge
Uzu (Vortex)	The stage in which a new coherence (order) arises spontaneously from within the tension
Taba (Bundle)	The stage in which the form is fixed and stabilizes as a reusable structure

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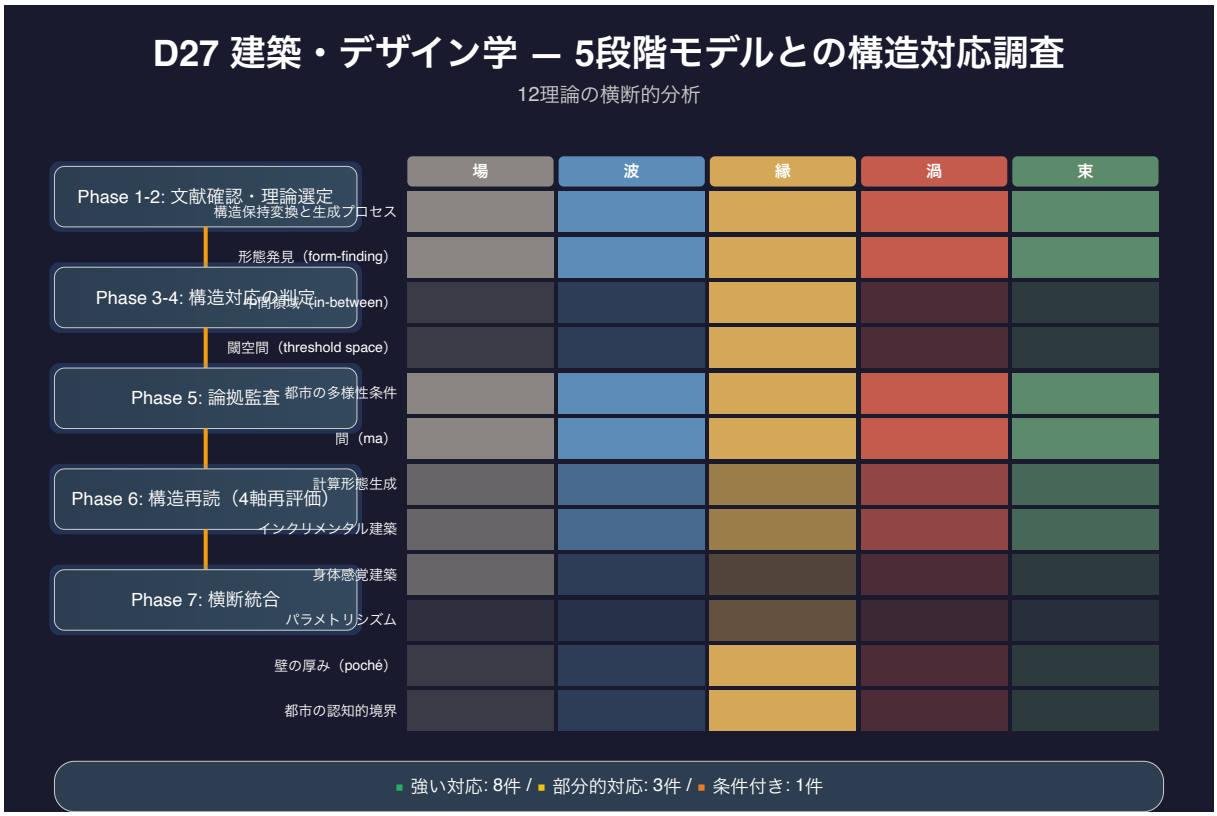

Figure 1: Domain — Research Overview Infographic

Overview of Structural Correspondences

Confidence	Theory group	Positioning
Near-established	Alexander (structure-preserving transformation), Otto (form-finding), van Eyck (in-between), Hertzberger (threshold space), Ma (Japanese spatio-temporal concept)	Correspondence with all 5 stages is clear, with particularly thick description of En
Compelling	Jacobs (urban diversity), Menges (computational morphogenesis), Aravena (incremental architecture), poché (wall thickness), Lynch (cognitive boundaries of the city)	Correspondence exists, but interpretive latitude remains in reading as process
Conditional	Pallasmaa (bodily-sensory architecture)	Partial correspondence with Ba and Wave only. Correspondence with En, Uzu, Taba is weak
No correspondence	Schumacher (parametricism)	"Continuous differentiation" moves toward erasing boundaries and is structurally in contradiction with En, the core concept of the 5 stages

Figure 2: Domain — Theory x 5-Stage Correspondence Matrix

Key Entry 1: Alexander's Structure-Preserving Transformation and Unfolding

- Alexander described the order of architecture and cities as "the configuration and interrelationship of centers," and called the process by which this order is generated step by step unfolding. Synthesized in his main work *The Nature of Order* (2002-2005) in 4 volumes, the theory is a development from *A Pattern Language* (1977). "Centers" here are not geometric center points but regions that come into being with intensity within a field.
- **As a matter of fact**: Alexander understood "life" as the degree of order common to architecture, cities, and nature, and explained its skeleton through the mutual support of centers = wholeness. The 15 properties (Levels of Scale, Strong Centers, Boundaries, Deep Interlock and Ambiguity, etc.) are a group of features repeatedly seen when a living wholeness arises. Unfolding as a generative process starts from the existing whole, and is a step-by-step process of adding structure-preserving transformations that strengthen without breaking the whole. The criterion for each step's selection is "whether the deep feeling of the whole increases."
- **As an interpretation of structure**: Here we read the repeating structure of transformations that strengthen order step by step while preserving the existing whole. The level of similarity is process, with particular attention to the order of "strengthening one step at a time without breaking the whole." Also, among the 15 properties, Boundaries (outlines with thickness) are defined as bands that strengthen centers, and the structural feature is read that the boundary is not a line but a region with generative power.
- **As an interpretive reading**: The step-by-step generative process of unfolding corresponds structurally to the whole of the 5-stage model. The state in which centers come into being as regions with intensity within Ba corresponds to Ba; the arising of differences among the 15 properties (Contrast, Alternating Repetition) corresponds to Wave; Boundaries and deep interlock (outlines with thickness and interlocking) correspond to En; the cohesion of Strong Centers (the arising of coherence as living centers) corresponds to Uzu; and the accumulated repetition of unfolding corresponds to Taba. In particular, the "strengthening without breaking" transformation directly corresponds to an operation that maintains the consistency of the whole at each step of the 5 stages.
- The difference in stage counts between the 15 properties (15) and the 5 stages (5) represents different angles on the same phenomenon — features and process — with only the labels of the stages differing. However, when directly mapping the 15 properties onto the 5 stages, care is needed not to confuse "feature description" and "process description."

Key Entry 2: Frei Otto's Form-Finding

- Frei Otto established a method of discovering the "good form" in lightweight structures not through the arbitrary choices of the designer but through the search for the equilibrium form created by material, mechanics, and boundary conditions. He used physical models of soap films (minimal surfaces), hanging chains, fabric and cable nets as iterative design tools, and developed integrated research practice in architecture, engineering, and materials science at the IL (Institute for Lightweight Structures) at the University of Stuttgart.
- **As a matter of fact**: Form-finding is a process of defining boundary conditions (span, support points, loads, materials), creating physical analogs, observing and measuring equilibrium forms, iteratively adjusting, and ultimately proceeding to engineering verification and implementation. The moment when a soap film takes a minimal surface, and the process in which a hanging chain traces a catenary, function not as finished-form models but as search tools through which forms are "found."
- **As an interpretation of structure**: Here we read the structure in which the design of boundary conditions governs form. The level of similarity is process, with particular attention to the undecidedness that "form is not fixed until boundary conditions are given" and the determining structure in which form "becomes fixed" within the combination of boundary conditions. Form-finding differs from other architectural theories in that form is generated as physical self-organization without the mediation of consciousness or intention.
- **As an interpretive reading**: Form-finding corresponds to the physical self-organization version of the 5 stages. The undetermined physical system (membrane and environment) is Ba; tension and fluctuation (mechanical instability, minute vibrations of the soap film) are Wave; boundary conditions (the frame, support points, and constraints that determine form) are En; the equilibrium form (a coherence arising as minimal surface) is Uzu; and the accumulation of the iterative design

process and the institutionalization of knowledge as the IL research report series are Taba. The finding that "boundary conditions determine form" provides a methodology for treating En as a set of manipulable variables.

Key Entry 3: van Eyck's In-Between

- Aldo van Eyck criticized the binary oppositions of modern architecture (interior/exterior, public/private, individual/collective, etc.) and argued that the in-between that mediates them is precisely what restores relationships and creates humanized space. Active in the context of Team 10, he proposed the doorstep as a unit of planning.
- **As a matter of fact**: For van Eyck, the boundary is not a line but "a domain called the doorstep," with thickness and activity. The idea is to reconcile opposing terms by "designing the in-between rather than joining them." His Amsterdam playground project (700-plus locations) is a practical example, inserting domains of children's activity into urban marginal spaces.
- **As an interpretation of structure**: Here we read the structure in which a thick domain is inserted between binary oppositions, and within it new relationships and activities occur. The level of similarity is structure, with particular attention to the point that the operation of "designing the in-between" recursively realizes the same structure at different scales — from the doorstep (micro) to the playground (meso) to the city (macro).
- **As an interpretive reading**: In-between is the most direct architectural-theoretical formulation of En in the 5-stage model. The existing binary oppositional spatial order is Ba; the manifestation of opposition (the disconnection of interior/exterior and public/private experienced as such) is Wave; the in-between / doorstep (the thick domain designed between the two terms) is En; the arising of humanized space (the appearance of a place where children play and people meet) is Uzu; and the institutionalization as a unit of planning (repetition of 700-plus playground projects, consolidation as design vocabulary) is Taba. Van Eyck's originality lies in presenting En not as an analytical concept but as a design operation.

Cross-Cutting Patterns

- The most recurring discovery across architecture is that "boundaries are not lines but bands"
- Second, architecture has a unique contribution distinct from other academic domains in treating En not as an analytical concept but as a design operation
- Third, the understanding of Ba is expanded
- Fourth, the ambivalence shown by Lynch's edge concept — "also a barrier, also a seam" — is a city-scale confirmation that En simultaneously contains both the potential for generation and destruction

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Unresolved Questions

- Alexander's "strengthening without breaking" transformation and van Gennep/Turner's anthropological "separation → liminality → incorporation" (breaking once and then reconstructing) connect to the fundamental question of whether Wave in the 5 stages is seen as "destruction" or "transformation." Within architecture itself, Alexander (the transformation camp) and Jacobs (the description of the destructive consequence called "border vacuum") illuminate this question from different angles.
- The concept of Ma was evaluated as limited to architectural spatial theory, but it is a concept that also spans performing arts, aesthetics, and traditional knowledge. An integrated positioning will be needed after evaluation in each domain is complete, and the risk of "symbolization of Japaneseness" needs to be managed independently in each domain.
- Whether the process by which algorithms search for optimal solutions in Menges's computational morphogenesis and the process by which physical laws generate equilibrium forms in Otto's work are different scale manifestations of the same generative structure, or essentially different processes, has not yet been resolved. The distinction between "computational optimization" and "spontaneous generation" is a question relating to the boundary conditions of the 5-stage model's scope.

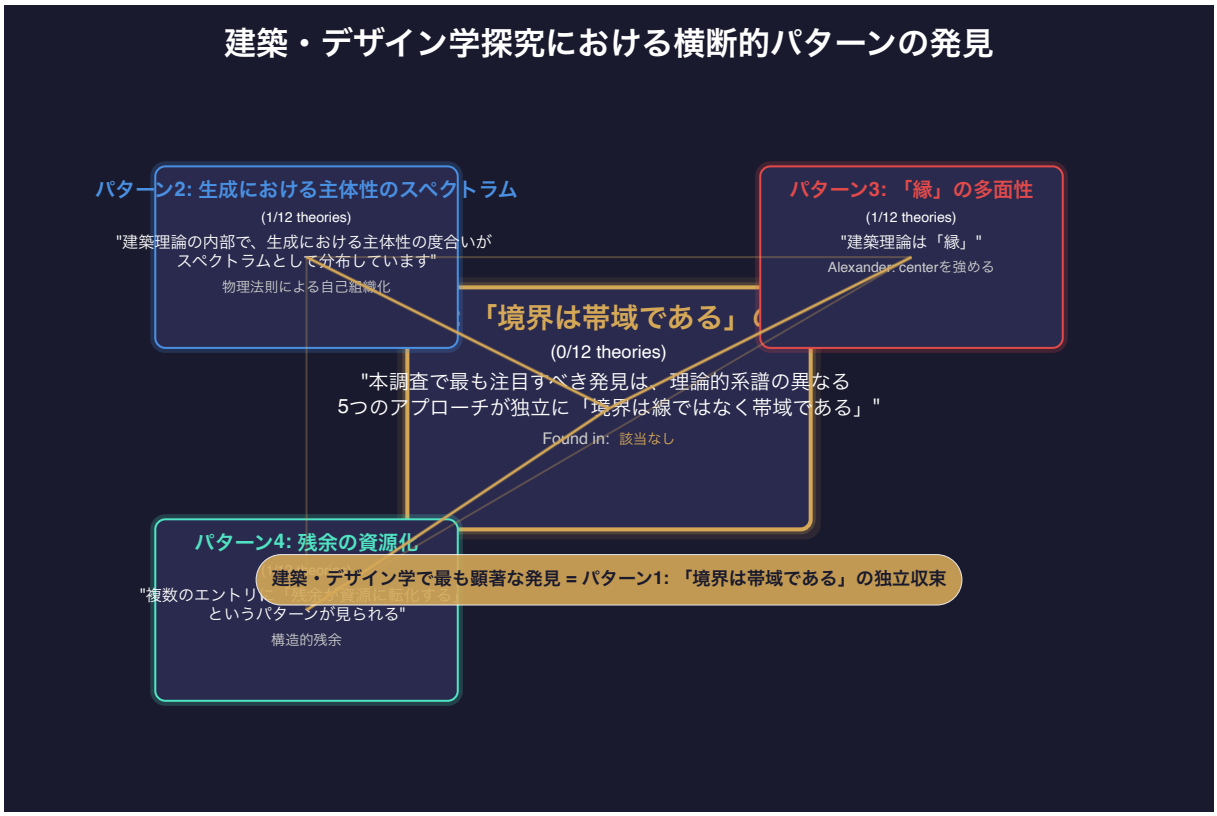

Figure 3: Domain — Cross-Cutting Patterns Diagram

- Schumacher's parametricism aims at "continuous differentiation," but this moves toward erasing boundaries and is judged to be structurally in contradiction with En, the core concept of the 5 stages. However, the possibility remains that "smooth differentiation" and "turbulence containing opposition" are different positions on the same spectrum, and there is room for future consideration.

Conclusion

- This survey confirmed that architecture is a domain with overall compelling structural similarity to the 5-stage model
- Near-established are the 5 theories of Alexander (structure-preserving transformation and unfolding), Otto (physical form-finding), van Eyck (in-between), Hertzberger (gradient design of threshold space), and Ma (architectural description of Japanese spatio-temporal concepts)
- There are two greatest insights to be gained from architecture